

UNIVERSITY OF BRISTOL

MSC BIOINFORMATICS

Evaluating Evapotranspiration Generalisation Under a Changing Climate using Machine Learning

Review and Plan by

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ABSTRACT (The abstract has a maximum of 200 words)

- Summary of background on ET/ML methods and FLUXNET
- Overall aim of the project
- Proposed approach for the project

Key words: Machine Learning, FLUXNET, Evapotranspiration.

Word count: 3000

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LITERATURE REVIEW (~ 1200 WORDS)

INTRODUCTION (~ 500 WORDS)

Evapotranspiration

- Overview of evapotranspiration (ET):

$$ET = Transpiration + E_{Canopy} + E_{Soil} \quad (1)$$

- Drivers of ET e.g Leaf Area index (LAI), Vapour pressure deficit (VPD), precipitation, air temperature etc.
- Importance and applications of ET predictability

Leaf Area Index

- What is LAI?
- How is this related to ET?

FLUXNET

- Overview of eddy covariance and how flux towers collect data
- Brief look at FLUXNET and how sites are grouped (region and vegetation types)

The Changing Climate

- Overview of climate change and effects on carbon dioxide levels
- How these could link to factors such as changing ET

CONTEMPORARY ET PREDICTION METHODS (~ 450 WORDS)

ML methods

- Brief overview of ML methods
- Performances of different models in the literature

KNOWLEDGE GAPS AND PROJECT SCOPE (~ 250 WORDS)

Knowledge Gaps and Unresolved Questions

- Findings from literature
- Address gaps in the literature
- Suggest areas for future efforts (which are not this project) - ?

Project Scope

- Suggest the area/question this project is going to address
- Include why this is relevant currently/addressable now
- "Do predictive ET ML models trained on historic FLUXNET data have a loss of performance under new climate conditions?"

RESEARCH PROJECT PLAN (~ 1500 WORDS)

OVERALL PROJECT AIM (~ 200 WORDS)

- Overview of how this project will answer the question/area proposed at end of introduction.

METHODS OVERVIEW (~ 1300 WORDS)

Data Acquisition and Selection

- How the data will be accessed and stored (FLUXNET).
- Time period (requires long term sites) and geographical location of data selected for modelling.
- Why these were selected and appropriate? (in terms of location/times, not specific structure).

Feature Engineering

- Selection of data features for use in training and why.
- How these were removed or adapted
- Why are these features appropriate?

ML Model Setup and Training

- How the data will be accessed and stored (remote sensing and FLUXNET).
- Explanation of model(s) selection (type of ML method)
- e.g If including LAi vs. not with other parameters kept identical across 2 models.
- Overview of model parameters (- might be too much detail?) and how each are trained.
- Why are these model choices/parameters appropriate for this task?
- Target features: FLUXNET latent heat flux (ET approximation)
- Predictor features: Meteorological data (from FLUXNET and others like LAi)

ML Model Evaluation

- Testing methods on unseen/held out historical (same as training e.g 5-20 years old) validation sets
- Comparison to other predictive models e.g in FLUXCOM
- Overall evaluation of model predictability on historical data

ML Model Predictions

- Predictions made on unseen and more recent data (e.g last 5 years)

- Overall evaluation of model predictability on recent data

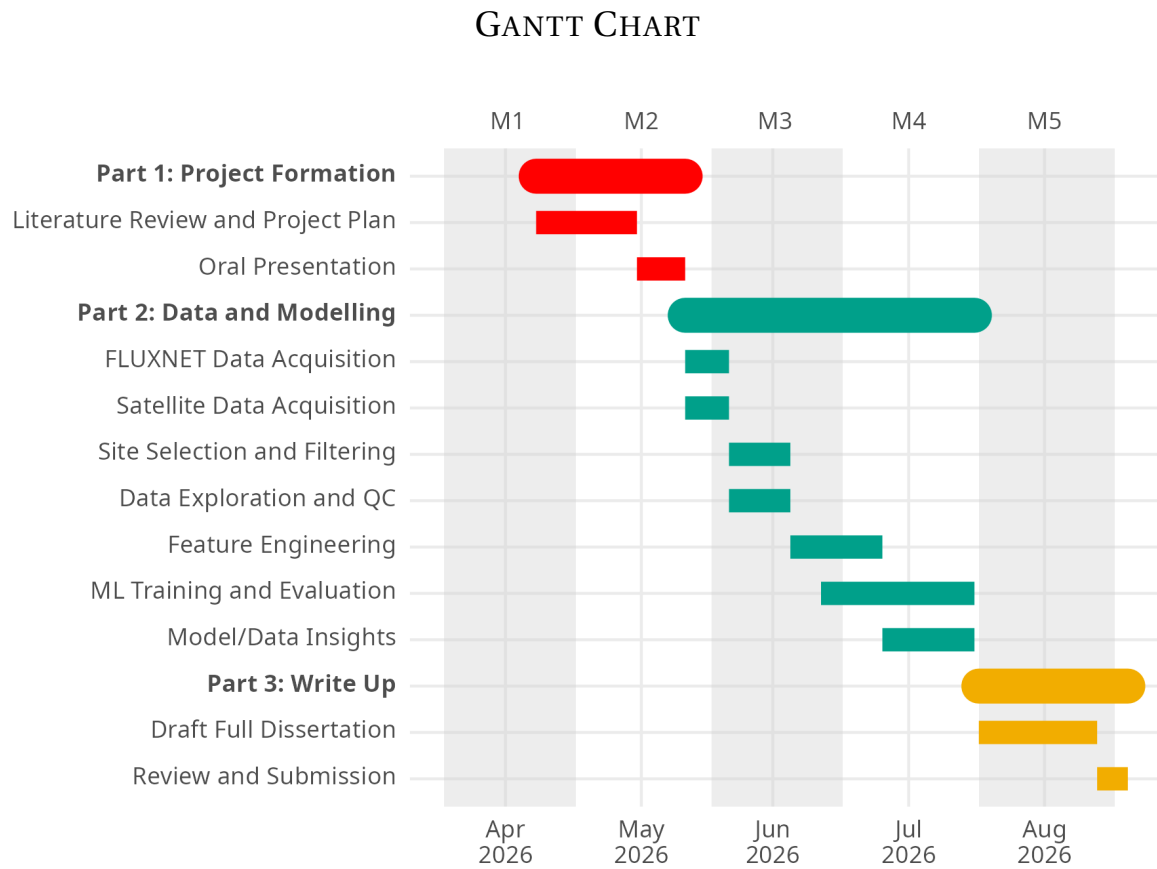


Figure 1: A proposed Gantt Chart for the duration of this project.

RISK REGISTER (~ 300 WORDS)

- 4-5 risks at about 60 words each
- Data access issues, overfitting, data gaps etc.

REFERENCES